CODING5S ECOSYSTEM | PEDAGOGY QUICK READ

AI Mentor Swarm & Behavioral Layer

Pillar 2 of the Coding5s Framework — Governing how AI assistance is delivered during guided practice

"Learn With AI. Don't Outsource Your Thinking."

EXECUTIVE SUMMARY

Pillar 2 is the **behavioral layer** of the Coding5s Framework. While Pillar 1 defines **what** the learner practices and how the five-stage progression is structured, Pillar 2 defines **how the AI should behave** while supporting that practice.

This layer is composed of specialized **Mentor Prompts** — behavioral specifications that apply **Controlled Cognitive Friction**, adapt guidance to the task at hand, and reduce the risk that AI assistance replaces the reasoning the learner is expected to perform. The objective is not to make the AI unhelpful, but to calibrate **the right amount of help for the cognitive task being practiced.**

1 WHAT A CODING5S MENTOR DOES

General-purpose AI assistants are optimized to be helpful and may provide complete solutions even when doing so weakens the intended learning task. Coding5s Mentors introduce behavioral constraints that change how assistance is delivered. A mentor may:

- Ask the learner to explain their current reasoning
- Request missing context before proceeding
- Provide diagnostic questions instead of immediate fixes
- Reveal hints progressively
- Critique an attempted solution
- Surface relevant constraints
- Adapt explanation depth to the learner
- Challenge design decisions or require a defended conclusion

2 COMMON MENTOR COMPONENTS

Individual mentors may use different structures, but five patterns recur across Pillar 2 specifications.

01 Role & Communication Style

A mentor can adopt a specialized professional role or recognizable conversational style to make interactions easier to understand and more engaging. **Personality is optional** — the behavioral objective matters more than theatrical presentation.

02 Context Adaptation

Mentors adapt to available information: language/domain, learner level, current stage, submitted artifact, learner explanation, and output language — based on **supplied or reasonably inferable context**, never invented assumptions.

03 Input Gates

When an exercise requires learner reasoning, a mentor may request the learner's interpretation before providing deeper assistance — preserving active participation rather than blocking unnecessarily.

Artifact Submitted → Explanation Missing → Ask What They Think → Guided Analysis

04 Assistance Boundaries

Coding5s does not impose a universal "no code" rule. A mentor may provide examples, scaffolding, broken code, partial implementations, diagnostic hints, documentation guidance, or review feedback. The constant boundary: **never perform the cognitive task the learner is currently expected to practice.**

05 Structured Interaction

Mentors may use consistent response structures when they improve learning: diagnostic checkpoints, progressive hints, comparison tables, reasoning prompts, review scorecards, reflection questions, or compact state summaries. **Formatting is a tool, not an invariant** — a mentor should not force complex structures when a simpler interaction works better.

3 INCLUDED MENTOR ARCHETYPES

Early reference implementations demonstrating different behavioral patterns within the Mentor Swarm.

STAGE 01 Legacy Architect — `legacy_architect.md`

OBJECTIVE
Help learners understand unfamiliar existing code before modifying it — reverse engineering & brownfield reasoning.

AI PERSONA
Systems archaeologist guiding structural discovery.

KEY STUDENT TAKEAWAYS
- Mapping structural relationships
- Identifying risky areas
- Discussing code smells
- Exploring consequences of changes
- Building a mental test plan before editing

STAGE 02 Concept Illuminator — `bridge_mentor_s1.md`

OBJECTIVE
Paradigm Bridge — Part 1. Introduce the conceptual differences between the source and target programming paradigms.

AI PERSONA
Conceptual translator / bridge builder.

KEY STUDENT TAKEAWAYS
- Conceptual mappings between paradigms
- Guiding analogies
- Partial scaffolding
- A structured first attempt

STAGE 03 Sparring Partner — `bridge_mentor_s2.md`

OBJECTIVE
Paradigm Bridge — Part 2. Reviews the learner's attempted translation and challenges assumptions inherited from the prior paradigm.

AI PERSONA
Critical sparring partner.

KEY STUDENT TAKEAWAYS
- Goes beyond right/wrong marking
- Recognizing why an old mental model doesn't map cleanly to the new paradigm

STAGE 04 Abstraction Architect — `math_calculus_architect.md`

OBJECTIVE
Mathematics, calculus, and geometric reasoning — conceptual and visual understanding before mechanical manipulation.

AI PERSONA
Visual & computational guide.

KEY STUDENT TAKEAWAYS
- Diagrams and computational experiments
- Python visualization & numerical examples
- Guided conceptual questions
- Connects symbolic operations to the relationships they represent

4 MENTORS ARE BEHAVIORAL COMPONENTS

A Coding5s Mentor is not necessarily a complete curriculum. Conceptually, mentor behavior sits between the learning task and the learner's own action:

Pillar 1
Learning Task + Stage Context
→
Pillar 2
Mentor Behavior
→
Interaction
AI-Assisted Interaction
→
Outcome
Learner Reasoning + Action

The same mentor behavior may support multiple languages, libraries, or domains when its constraints remain relevant. Likewise, the same lesson can use different mentor behaviors depending on the learning objective.

5 BUILDING NEW MENTORS

New mentor archetypes can target specific forms of reasoning. A useful specification should clearly document seven elements:

#	ELEMENT	WHAT IT ANSWERS
1	**Target Task**	What kind of reasoning does it support?
2	**Expected Inputs**	What should the learner provide?
3	**Behavioral Rules**	How should the AI respond?
4	**Assistance Boundary**	What should the mentor avoid doing for the learner?
5	**Adaptation Rules**	What context can change its behavior?
6	**Known Limitations**	Where does the prompt behave inconsistently?
7	**Example Interactions**	What does successful use look like?

Code Review · Debugging · Architecture · Performance · Security · Accessibility · Data Analysis · Tech Communication · Troubleshooting · Paradigm Transition

6 TESTING MENTOR BEHAVIOR

Mentor prompts should be tested through realistic interactions rather than judged only by reading the prompt. Useful observations include:

- ✓ Whether the mentor preserves the intended assistance boundary
- ✓ Whether hints become too revealing
- ✓ Whether formatting remains useful
- ✓ Whether it asks unnecessary questions
- ✓ Whether instructions degrade during long interactions
- ✓ Whether behavior changes significantly across models

Cross-model testing is encouraged, especially when claiming a mentor is model-agnostic. Model behavior can change over time, so mentor prompts should be treated as **behavioral specifications that may require iteration**, not permanent guarantees.

7 HOW TO USE A MENTOR

1. Open the desired mentor **.md** file.
2. Copy the mentor specification.
3. Provide it through the most appropriate instruction mechanism available in the selected AI platform.
4. Supply the requested learner input.
5. Interact with the mentor while performing the actual learning task.

Exact system-prompt, custom-instruction, project, or conversation setup will depend on the AI platform being used.

8 RELATIONSHIP TO THE OTHER PILLARS

Pillar 1
What should the learner practice? How does difficulty progress?

Pillar 2
How should AI behave while supporting that practice?

Pillar 3
What useful context should persist as the learner progresses?

Pillar 2 can also be used independently wherever an AI interaction benefits from explicit behavioral boundaries and learner-centered guidance.

9 THE SWARM IDEA

The term **Mentor Swarm** does not require many AI agents to run simultaneously. It describes a collection of specialized mentor behaviors that can be selected according to the learning task: One Learner → Different Cognitive Problems → Different Mentor Behaviors. Future implementations may orchestrate multiple mentors together, but the core value of Pillar 2 already exists when a single specialized mentor changes how AI assistance is delivered.

> ***"The mentor should help the learner think better — not simply think for them."***

i TECHNICAL FOOTER

KEY ARCHITECTURAL PILLARS

- **Pillar 1 — Practice Structure**
  Defines the five-stage learning progression and what the learner practices at each stage.
- **Pillar 2 — Behavioral Layer**
  The Mentor Swarm — governs how AI assistance is delivered during that practice (this document).
- **Pillar 3 — Persistent Context**
  Defines what useful context should carry forward as the learner progresses.

SYSTEM GUARANTEES

- **Controlled Cognitive Friction**
  Assistance is calibrated by design, not maximized by default.
- **Task-Appropriate Boundary**
  No blanket "no code" rule — the boundary changes with the learning task.
- **Grounded Context Adaptation**
  Adapts only to supplied or reasonably inferable learner context.
- **Testable Specification**
  Every mentor is treated as an iterable behavioral spec, validated across models.

REFERENCES

- **Legacy Architect**
  `legacy_architect.md`
- **Concept Illuminator**
  `bridge_mentor_s1.md`
- **Sparring Partner**
  `bridge_mentor_s2.md`
- **Abstraction Architect**
  `math_calculus_architect.md`